15 Three solid shapes **A**, **B** and **C** are similar.

The surface area of shape **A** is $4\,cm^2$
The surface area of shape **B** is $25\,cm^2$

The ratio of the volume of shape **B** to the volume of shape **C** is 27 : 64

Work out the ratio of the height of shape **A** to the height of shape **C**.
Give your answer in its simplest form.

A : B lengths = √4 : √25 = 2 : 5

1 mark
Square roots of the area ratio

B : C lengths = $\sqrt[3]{27} : \sqrt[3]{64}$ = 3 : 4

1 mark
Cube roots of the volume ratio

	A	B
×3	2	5
×3	6	15

	B	C
×5	3	4
×5	15	20

1 mark
Match the B values

A : B : C = 6 : 15 : 20

A : C = 6 : 20 = 3 : 10

1 mark
Simplify the requested ratio

3 : 10

(Total for Question 15 is 4 marks)

16 Prove algebraically that $0.2\dot{5}\dot{6}$ can be written as $\frac{127}{495}$

(Total for Question 16 is 3 marks)

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